

Md Shahid Ali Khan

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PROFILE SUMMARY

Software Engineer with **3+ years of experience** in **Java and Spring Boot**, developing **microservices and distributed systems** using **Kafka, Redis, REST APIs, concurrency, and asynchronous processing** in the financial services domain.

TECHNICAL SKILLS

Languages & Core CS: Java, Data Structures & Algorithms (DSA), OOP, Operating Systems, DBMS

Backend & Distributed Systems: Spring Boot, Spring Data JPA, Hibernate, REST APIs, Microservices, Apache Kafka, Redis, Caffeine, Concurrency, Multithreading, Resilience4j, System Design

Databases, Analytics & Cloud: PostgreSQL, MySQL, ActivePivot, MDX, AWS (Lambda, S3, EC2, CloudWatch)

DevOps & Tools: Git, GitHub Actions, CI/CD Pipelines, Docker

WORK EXPERIENCE

Societe Generale

Software Engineer

Bangalore, India

July 2023 – Present

Financial Risk Reporting Platform | *Java, Spring Boot, Microservices, REST APIs, Kafka, ActivePivot, Caffeine, PostgreSQL*

- Developed and enhanced **Java/Spring Boot backend services and REST APIs** for an enterprise financial risk analytics platform supporting **NPV, NIM, GAP, sensitivity analysis, and regulatory reporting**.
- Developed analytical reporting workflows using **MDX** against **ActivePivot in-memory OLAP cubes**, leveraging **olap4j/qfs-olap4j** for Java-based OLAP connectivity and **RafalQuery APIs** for analytical data retrieval, supporting **50K+ record** report workloads across financial risk analysis and reporting.
- Implemented **Caffeine-based L1 caching**, optimizing cache keys, TTLs, and eviction policies to reduce API response time from **10s to under 2s** and report generation time from **2min to 30s**.
- Refactored long-running report generation into **Kafka-driven asynchronous workflows**, enabling concurrent report processing and improving API responsiveness.
- Implemented resilient **Kafka consumer workflows with retry handling** and **MessengerClient** consumers for automated email notifications on failed report batches.
- Implemented scheduled background jobs and **dedicated thread pools** for report, export, and cache-refresh workflows, with **Resilience4j Circuit Breaker** for failure handling.
- Contributed to **Angular feature development and production bug fixes**, supporting GitHub Actions CI/CD.
- Conducted **code reviews and knowledge-transfer sessions**, mentoring 2 interns who subsequently converted to FTE.

Societe Generale

Software Developer Intern

Bangalore, India

March 2023 – July 2023

Rejected Transaction Impact Analysis & Script Migration

- Built a **Spring Boot**-based analysis platform to categorize rejected transactions by risk, processing **100K+ transaction records** using optimized SQL queries to accelerate remediation workflows.
- Migrated **70+ legacy Unix shell scripts** to Linux and refactored workflows, improving system compatibility, maintainability, and operational efficiency.

PROJECTS

Distributed Workflow & Job Processing Platform | *Java, Spring Boot, PostgreSQL, Kafka, Redis, Resilience4j, Docker*

- Built an asynchronous **job-processing platform** with persistent job lifecycle management, atomic state transitions, Kafka consumer groups, and concurrent workers.
- Implemented the **Transactional Outbox pattern** for reliable DB-to-Kafka event propagation, preventing dual-write inconsistencies during failures.
- Built **Caffeine + Redis L1/L2 caching** using the Cache-Aside pattern, along with Redis-based **distributed rate limiting** using atomic Lua operations.
- Engineered **at-least-once Kafka processing** with acknowledgements, offset management, atomic job claiming, and idempotency to handle duplicate message delivery.
- Implemented resilient job execution with **exponential backoff, nextAttemptAt retries, Kafka error handling, and Dead Letter Topics (DLT)**, separating job and messaging failures.
- Added **Resilience4j Circuit Breaker** for Redis dependency failures and **Redis distributed locking** with expiry for coordination across application instances.

EDUCATION

Birla Institute of Technology, Mesra (BIT Mesra)

B.Tech in Information Technology | GPA: 8.60/10

Ranchi, India

2019 – 2023

- Solved **650+ Data Structures & Algorithms problems**, demonstrating strong problem-solving and algorithmic skills.
- Secured **National Rank 844** in CODEKAZE 2021 among **200,000+ participants**.